M1: Approach – Green Home Score & Retrofit Recommendation Engine

Megathon X: Singularity | Problem Statement 2 (Lloyds) | **Team: Commit(ment)-Issues** | **Team No. 78**

1 Problem statement and our understanding

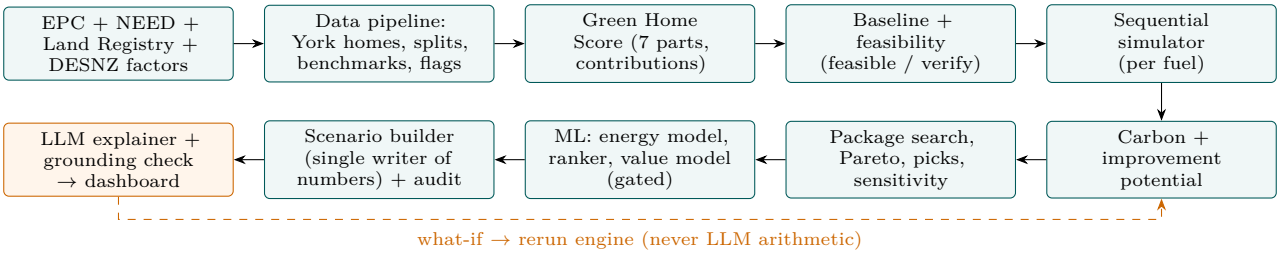
Homeowners want an energy-efficient home but do not know where to start. The system must answer: *How green is my home today, what should I improve, how much energy and carbon could I save, and could this increase my property's value?* It must (1) compute a 0–100 **Green Home Score** from seven fixed-weight parts (energy efficiency 25%, energy intensity 20%, insulation 15%, heating 15%, windows 10%, renewables 10%, smart controls 5%); (2) flag Poor/Very poor homes and rank them by **improvement potential**; (3) recommend **feasible** retrofits in an order that respects interactions (insulation before heat pump); (4) estimate savings **separately for electricity and gas** and convert them to carbon with documented factors; (5) give an **indicative** property-value range.

Our reading of the hard constraints: the LLM never calculates or invents a number—every figure comes from a documented model, and a what-if reruns the calculation first. A heat pump lowers gas but *raises* electricity, so per-fuel accounting (with negative savings) is essential. Missing data must stay missing (never a silent 0), large homes must not be penalised for total use (we use kWh/m^2), and historical price associations are not causal.

2 Idea / abstract

We build a **deterministic, auditable engine** on **real public data** for the City of York (65,872 homes, 95,348 EPC certificates), with an LLM only as the explanation layer. EPC records give each home's features and rating; DESNZ NEED gives metered-consumption benchmarks and *measured* retrofit savings; Land Registry gives sale prices; DESNZ 2026 factors give carbon. A sequential simulator applies retrofits in a fixed physical order on per-end-use energy (space heat, hot water, cooking, grid electricity), so overlaps are handled and every kWh reconciles. A package search enumerates all feasible retrofit combinations, keeps the Pareto-optimal ones (cost, carbon, energy) and presents three clearly-labelled picks (lowest cost and most carbon per pound, each cutting ≥10% of baseline emissions; largest carbon saving) with a generated decision audit. If no package qualifies, this is shown as such, never by lowering the threshold. Every value carries provenance (observed, benchmark, user-provided, modelled, assumption).

3 High-level approach and architecture



- **Score:** documented, monotonic sub-score maps (e.g. SAP capped 0–100; intensity $100\,(p_{90}-x)/(p_{90}-p_{10})$ with percentiles frozen on train only; element ratings 10/30/55/80/100). A missing part is scored 50 with a [0,100] range and completeness $k/7$; >2 missing $\Rightarrow$ score unavailable.
- **Feasibility:** catalogue rules (e.g. no loft for a flat with a dwelling above, no cavity fill after 2000, no PV on flats); a missing input gives *needs verification*, never *infeasible*. ML may only reorder feasible measures.
- **Simulator:** gas split into cooking (3.05%, untouched) and space/water (77/23)[1]; NEED impacts (% of total gas) converted to the space-heating end use; fixed order fabric → glazing → controls → heat pump (heat/SCOP 3.0 moves to electricity) → solar (grid import only). Reductions multiply.
- **Carbon:** $\Delta C = \sum_{f\in\{elec,gas\}} \Delta E_f \cdot EF_f$ (0.14395 / 0.18231 kg CO_2e/kWh, DESNZ 2026).
- **ML (in progress):** LightGBM energy model on NEED (replaces benchmarks only if it beats them), recommendation ranker on EPC assessor recommendations, hedonic value model on 4,000 matched sales reported as an *indicative* range (not causal), enabled only after a manual match-precision gate (≥0.90) and a held-out range-coverage check (75–85%).
- **LLM:** receives only the scenario JSON + audit and explains score, recommendations and trade-offs. It writes placeholders only; code fills in every number from the scenario result.

Status at M1: data pipeline, score, feasibility, simulator, carbon, package search, sensitivity, audit and scenario builder are implemented (169 passing tests, ∼1 ms per home). Demo: an EPC-E 1970s house scores **44/100 (Poor)**; its feasible package saves 20,957 kWh gas, adds 3,675 kWh electricity (heat pump), cuts purchased energy by 68% and carbon by 3.29 t/yr; cheapest ≥10% option: cavity-wall insulation (£1,000).

[1] DESNZ, *Energy Consumption in the UK 2026*, End-use data tables, Table U3 (domestic natural gas by end use), mean of 2021–2025: cooking 3.05%; space heating 77.3% of space + water heating gas.

4 Planned testing methodology

- **Leakage-safe splits:** property-level train/val/test (46,151 / 9,797 / 9,924); score percentiles and models fitted on train only; no field unavailable at prediction time; EPC rating at sale taken only from earlier certificates. Automated data checks (unique homes, split isolation, no zero-filled gaps, provenance).
- **Hand-calculated reference suite:** energy balance with no retrofit; 3% of 12,000 kWh = 360 kWh; loft+cavity overlap < sum; fabric-order invariance; full/partial/unknown heat pump (gas ↓, electricity ↑); solar applied last; infeasible measure blocked; zero baseline ⇒ unavailable; carbon signs. Demo homes are reproduced from independent hand arithmetic to 10^{-6}.
- **Property tests:** monotonicity of every sub-score, class boundaries (29/30, 49/50, 64/65, 79/80), weights sum to 1, per-fuel steps reconcile, same request ⇒ same scenario ID.
- **External validation:** simulated multi-measure packages vs NEED *measured* combination savings (tolerance ±5 pp). Our simulated combined savings **exceed** NEED's measured figures by 5.6–13.8 percentage points, outside tolerance. We report this rather than tuning it away; likely causes include differences in home mix and the gap between modelled and real behaviour. Single-measure NEED figures are model inputs, so they are not re-used as tests.
- **Decision quality:** on held-out homes, compare three policies (fixed checklist, greedy, full package search) under the same objective and budget (£5k, £15k, unlimited); report regret against the best package within budget and feasibility violations. This measures performance under our documented simulator, not real-world optimality.
- **ML:** validation metrics against simple baselines (benchmark medians for energy; rules-only ranking for recommendations; precision@k, MAE); value range coverage on held-out sales.
- **LLM:** held-out scenarios (missing/conflicting data, infeasible or interacting retrofits, electricity rising after a heat pump, insufficient data). The LLM writes placeholders only; code fills in numbers from the scenario result. A checker confirms no stray digits, that every placeholder exists, that the scenario ID matches, and that signs and directions are correct; any failure falls back to a template. Targets: groundedness ≥90%, factual accuracy ≥90%, unsupported claims <5%.
- **Robustness:** one-at-a-time sensitivity (baseline ±15%, SCOP 2.5/3.5, gas split, impact low/high) reporting whether recommendations survive; latency target <1 s per scenario.